Compiler Optimizations for Attention and Functional KV Caches

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Main result. On a Tesla T4, winner-cache factoring lowers a pretrained two-candidate step's IREE latency by **26.46% at context 2048**. Sparse synthetic cache forwarding reduces measured DRAM bytes by **99.92%**. These are different workloads; full-model DRAM reduction is not consistent.

1 Problem and baseline investigation

- **GQA and caching.** A KV cache stores earlier tokens' keys and values. With H_q query and H_{kv} KV heads, GQA shares each KV head across $g = H_q/H_{kv}$ query heads: $h_{kv} = \lfloor h_q/g \rfloor$ [1, 2].
- **Initial hypothesis.** Explicit `repeat_interleave`/reshape can expand K/V tensors. A C++ rewrite improved the original default CPU case **18.47 to 9.94 μs (1.86×)**, but stronger existing IREE settings remove the tested expansion too.
- **Deciding settings.** 20 fixtures, 160 compilations and 96 numerical checks established that both flags below were needed for the tested grouped cases; O3 alone was insufficient [13, 16].

```
--iree-opt-generalize-matmul
--iree-dispatch-creation-enable-aggressive-reshape-movement
```

The subsequent study targets missed functional-cache patterns.

2 Main transformations

Let C be an immutable history, U a t-row update at position p, and $C' = \mathrm{update}(C, p, U)$.

$$\underbrace{C'[i]}_{\text{read after update}} \longrightarrow \begin{cases} U[i-p], & p \le i < p+t, \\ C[i], & \text{otherwise.} \end{cases}$$

Read forwarding: sparse/gather consumers access the base or patch directly. An unobserved updated tensor may disappear; a returned full cache still needs its required representation.

$$\underbrace{\mathrm{select}(c, \mathrm{concat}(C, a), \mathrm{concat}(C, b))}_{\text{select full candidate caches}} \longrightarrow \underbrace{\mathrm{concat}(C, \mathrm{select}(c, a, b))}_{\text{select small rows, then commit}}.$$

Winner factoring: same history/append axis, scalar Boolean c. The model has **60 matches** (K/V in 30 layers); ordinary decode has **zero**. Candidate attention computations remain.

3 Implemented extensions and use cases

Transformation	Implementation / intended use	Boundary / prior work
Read/gather forwarding	Base-or-patch reads for forked sparse attention; normalize gather indices.	Established read-over-write algebra [3, 4]; full cache observations remain.
Winner commit factoring	Select small rows before a common-history append / equal-region update.	Custom branching step; standard selection algebra, not complete beam/speculative decoding.
Update coalescing	Merge adjacent small writes before forwarding multi-token chains.	Reject gaps, overlaps or observed intermediate versions. Classical update simplification.
Snapshot protection	Preserve returned old caches and supported borrowed-buffer ownership.	Correctness repair; sharing/COW precedents [5, 6]. Invalid outputs are excluded.
Projection / partial RoPE	$q(CW_K)^T = (qW_K^T)C^T$; project content separately from rotary coordinates.	MLA absorption / decoupled RoPE are prior art [7, 8]; require a numerical contract.
Native-attention guard	Preserve complete feature reduction before online softmax; decomposed fallback for failures.	Established online reduction [10, 11]; early upstream decomposition is a CPU alternative.
Stable trig reduction	Bounded FP64 argument reduction inside small FP32 sine/cosine kernels.	Repair for $32 < \lvert x \rvert \le 8192$, kernel size ≤ 4096. Not full Payne–Hanek or an all-input accuracy proof [9].

4 Compiler integration



Standalone MLIR 19 driver [12], with a standard-IR bridge to IREE 3.13/LLVM 24 before dispatch creation. Type, shape, SSA identity, indexing, use and lowering-state guards reject unsupported patterns. No new GPU kernel or complete patched in-tree build.

5 Evaluation protocol

Platform	Tesla T4 (`sm_75`), driver 580.82.07; FP32, TF32 off. Torch 2.11.0+cu130 references; IREE 3.13.0 (compiler `89ece87b8985`); MLIR 19 driver.
Model	SmolLM2-135M-Instruct [14], checkpoint `12fd25f77366`; 30 layers, 9Q / 3KV heads, width 64. Track B: custom two-candidate step. Track A: ordinary-decode no-match control.
Matched baseline	Both variants: O3, both shape flags, shared split-argmax workarounds, same trig repair. Set `--iree-input-demote-f64-to-f32=false`; disable numeric precision reduction in both.
Timing	Three contexts × three prompts × three repetitions of 31 samples. Random backend order, resident outputs, context synchronization. Median of nine repetition medians; compilation/prefill/upload excluded.
Traffic	Nsight Compute 2025.3.1 app-range replay [15]; prompt 0, three processes/backend/context, three fixed-input calls/range, bytes divided by 3. Separate unprofiled timing. 51 valid A + 61 B profiles; 112 CSV totals rechecked.

6 Final IREE model results

Context	Base ms	Opt. ms	Latency lower	Base GB	Opt. GB	Δ DRAM
2048	43.264	31.815	**26.46%**	2.417	2.323	−3.91%
3072	63.326	60.386	4.64%	3.093	3.132	+1.24%
4096	82.409	77.858	5.52%	3.881	3.870	−0.27%

Matched IREE comparisons; decimal GB/MB. ΔDRAM is signed read+write change (positive: more traffic). Timing: three prompts; counters: prompt 0.

- **Correctness:** 9 natural + 18 forced-winner cases pass per IREE B variant, with exact old-cache prefixes and unchanged inputs. Worst error: 5.05×10^{-5}; fixed reference tolerance: atol= 5×10^{-4}, rtol= 3×10^{-3}.
- **Accuracy control:** unmodified IREE passed 5/9 natural cases; trig repair enabled 9/9. Both timed variants receive that repair.
- **Mechanism:** writes fall but reads rise. At context 2048: 987→1047 kernels/call; 188.790→188.928 MB logical memcpy/call. Full-model copy elimination is not demonstrated; memcpy payload is not DRAM.

7 Synthetic mechanism and controls

Track B fixture	Base ms	Opt. ms	Speedup	DRAM MB: base → opt.
Small sparse forks	0.288	0.246	1.17×	5.831→0.128 (97.80% less)
Large sparse forks	1.777	1.230	1.44×	178.897→0.137 (99.92% less)
Dense no-rewrite control	0.545	1.268	**0.43**×	0.285→0.275 (small / variable)

Track A large fixture: 178.228→0.132 MB (99.93% less), a second same-GPU-family session. Synthetic FP64 NumPy tolerance: atol= 5×10^{-5}, rtol= 2×10^{-3}.

8 CPU extensions and limitations

Earlier CPU study	Recorded gain	Scope [16]
Latent projection placement	3.93× geometric mean	16 synthetic decomposed cases; strong unabsorbed baseline, not optimized MLA serving.
Cache/block and winner cases	2.39×/7.90×;3.60×	Two synthetic blocks and one winner-attention case; separate operator study.
Coalescing; snapshot reads	2.17–2.34×;2.58×	4–16-token chains; fastest correct snapshot baseline. Regressions retained.

- **Evidence strength differs:** logical elimination (IR), allocation requests (runtime), DRAM bytes (Nsight) and latency (separate timing) are different claims.
- **Timing uncertainty:** the no-rewrite regression and changing GPU clocks limit causal attribution. No controlled cross-session confidence interval or exact kernel cause is established.
- **Coverage:** one FP32 small model, one GPU family, custom branching. No general serving-throughput, all-GQA, mixed-precision or global-novelty claim.
- **Negative cases:** stock lowering mutates inputs in a single-fork probe; it is excluded. Wide-Q/K numerical failures and irregular masked CUDA compile failures remain recorded.

Conclusion. Selective functional-cache handling improves the measured IREE branching step. Sparse synthetic forwarding reduces DRAM traffic. Graph coverage and full-model traffic causality remain limited.

Attribution rule. GQA, MLA absorption and decoupled RoPE have identifiable originating papers below. Read-over-write has classical foundations. For selection factoring, adjacent-update merging and this exact compiler repair, no definitive first paper is established here. Cache-sharing papers are related systems precedents, not attribution for our precise MLIR patterns.

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